RiskPulse AI

Intelligent Financial Risk & Portfolio Stress Lab

From unstructured information to actionable financial risk intelligence.



Executive Risk Overview - live dashboard (synthetic data)

SIGNALS	HIGH IMPACT	TESTS PASS
254	56	93

S&P Global & Crisil Campus Hackathon 2026

Mohit Kumar Gautam · IIT Delhi

AI/NLP risk engine | Module B: stress testing | Offline · explainable

PROBLEM & APPROACH

Turning noisy text into risk decisions

The problem

1. **Volume**
   Risk-relevant news and posts arrive faster than analysts can read them.

2. **Ambiguity**
   Tone is not impact: a negative tweet about a logo is noise; a missed coupon is not.

3. **Disconnect**
   Text insight rarely reaches portfolio numbers: what does this event cost us?

Our approach: one explainable pipeline

1. **Ingest**
   News + social adapters, validation, dedupe, stable IDs

2. **Understand**
   Entities & tickers, finance-tuned sentiment, 10 event types

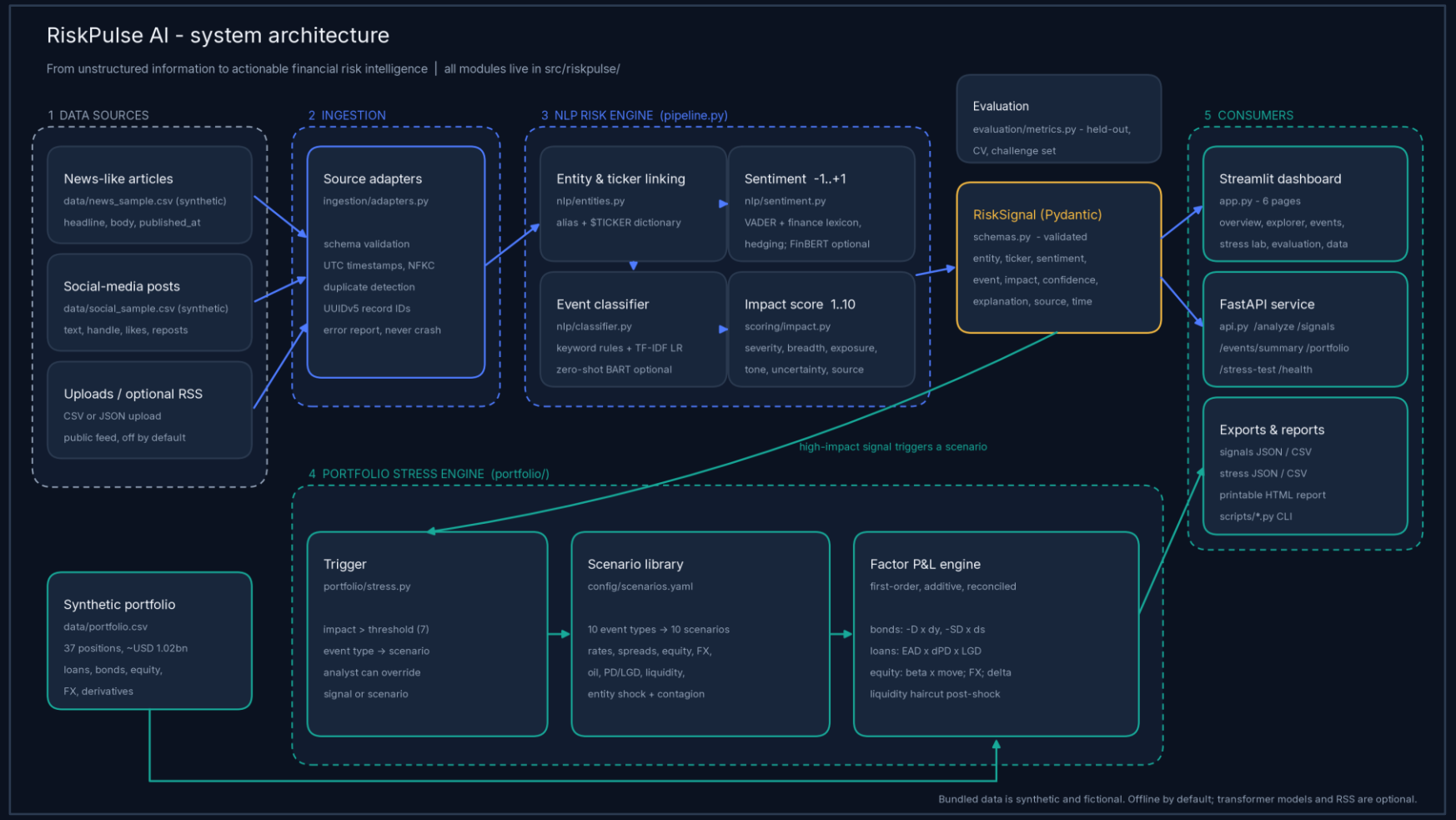
3. **Score**
   Impact 1-10 from transparent, configurable factors + explanation

4. **Stress**
   Impact > 7 auto-selects a scenario and revalues a 37-position book

Outputs: dashboard · REST API · JSON / CSV · printable report

ARCHITECTURE

System design: modular, offline, one engine everywhere



Tech choices

Python 3.11
pandas, NumPy

scikit-learn
TF-IDF + logistic regression

VADER
+ finance lexicon

Pydantic
validated signal schema

FastAPI
REST for downstream apps

Streamlit + Plotly
6-page dashboard

PyYAML
editable scenarios & weights

Optional
FinBERT, BART-MNLI

Dashboard, API, CLI and reports call the same riskpulse modules: displayed and exported numbers always agree.

RISK ENGINE

AI/NLP risk intelligence with a reason for every score

1

Sentiment -1 to +1

VADER + 116-term finance lexicon, phrase rules, hedge dampening; company names masked. 57.5% -> 70.4% accuracy vs naive VADER.

2

Event classification

Keyword rules (auditable evidence) + TF-IDF logistic regression over 10 categories; weak evidence -> 'Unclassified', close calls flagged ambiguous.

3

Impact score 1 to 10

Base severity + intensity + breadth + portfolio exposure + tone - uncertainty - source reliability. Impact is severity, not confidence.

4

Explanation

Every signal carries keywords, sentiment drivers, entity exposure and the arithmetic behind its impact score.

GENUINE OUTPUT - news record from data/news_sample.csv

"Meridian Shipping missed a coupon payment on its senior bonds. Rating agency downgrades Meridian Shipping to CCC and places it on negative watch ..."

Entity / ticker	Meridian Shipping (MRDS)
Event type	Credit Event (confidence 1.00)
Sentiment	-1.00 Negative
Impact	**9 / 10 -> stress trigger**
Source / time	news · 2026-10-01 13:40 UTC

Why impact = 9

base 6.5 + intensity 0.5 + exposure 0.8 + tone 1.5 = 9.3 -> rounded 9

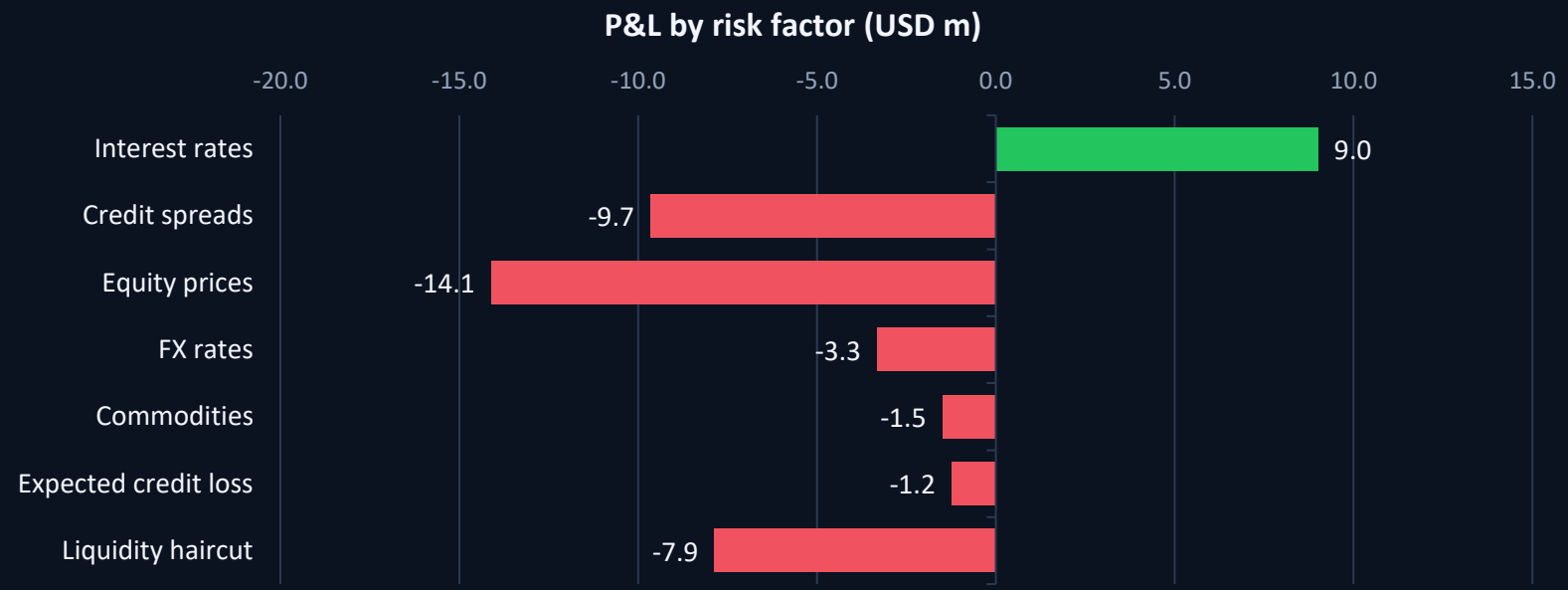
Keywords: default, downgrad, missed a coupon, restructur

MODULE B

Event-driven portfolio stress testing

Auto-triggered: Market / Liquidity signal (impact 10 > 7) -> scenario 'Market / liquidity stress', severity scale 1.25

BEFORE	AFTER	P&L	P&L %
$1.02bn	$990.41m	-$28.69m	-2.81%



Assumptions (explicit)

- Rates -50 bp (flight to quality)
- Equities -18.8% x beta
- Spreads +125 bp BBB / +312 bp HY
- Loan PD x1.25, LGD +3.8 pp
- Liquidity haircut after market moves
- Bonds: -ModDur x dy (bp / 10,000)
- Derivatives: delta / DV01 approximations

Second example (entity-specific)

MRDS credit event: -$12.47m (-1.22%); CDS hedge offsets $3.26m. Totals reconcile to the cent.

RESULTS

Results, validation and domain impact

AUTOMATED TESTS

93 / 93

ingestion, NLP, maths, API, UI

RECORDS INGESTED

266 -> 254

12 duplicates removed

GROUPED CV MACRO-F1

0.976

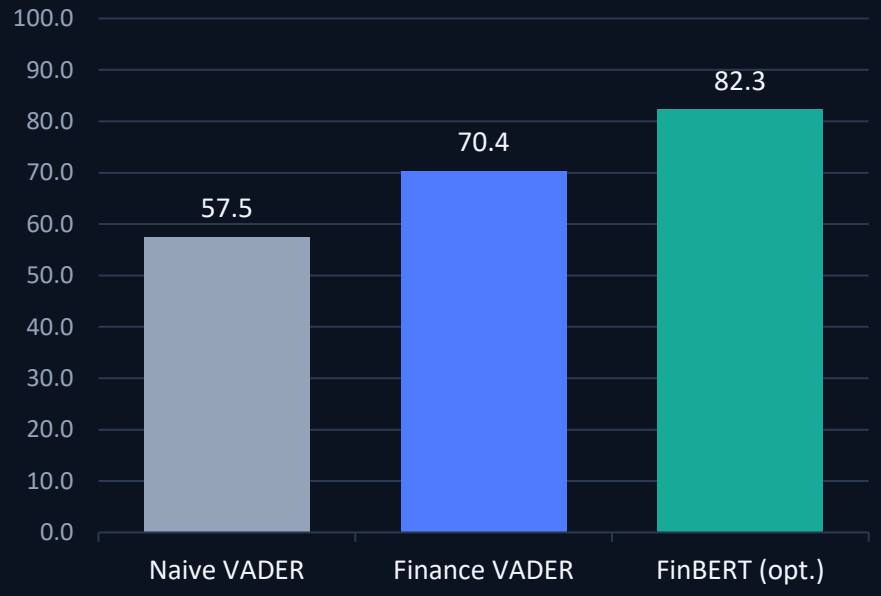
held-out templates (optimistic)

P&L RECONCILIATION GAP

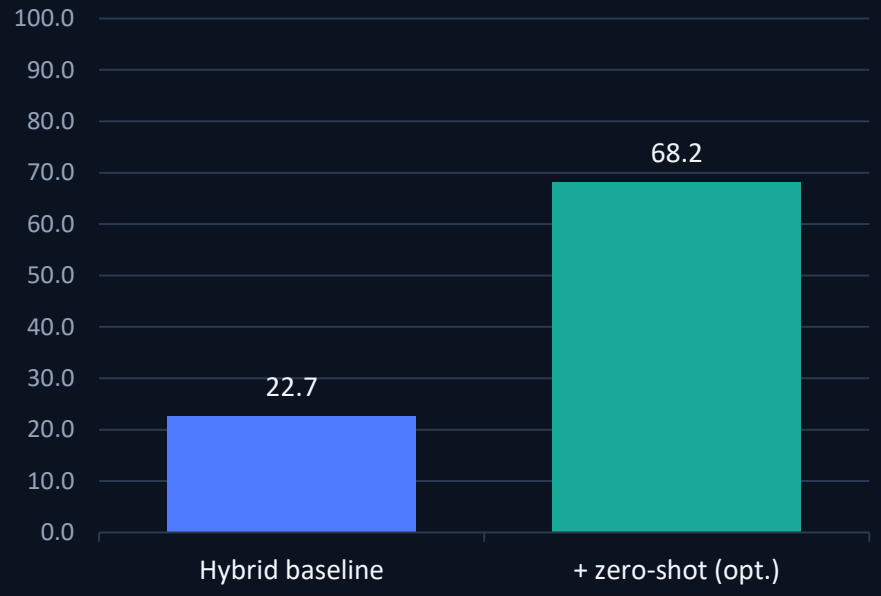
0.00

components = positions = total

Sentiment accuracy, % (628 labelled)



Event accuracy on 44 unseen paraphrases, %



Why it matters

- Early warning: minutes from headline to quantified exposure
- Explainable scores an analyst can audit and challenge
- Links NLP output directly to portfolio P&L by asset class
- API + CSV/JSON feed limits, alerts and credit reviews

All metrics measured on synthetic data by scripts/evaluate_model.py; they show the pipeline works, not real-market predictive power.

HONEST ASSESSMENT

Limitations and next steps

What remains unvalidated

- Synthetic, template-generated text; no real-market labels or price reaction data
- Baseline classifier generalises poorly to unseen phrasing (23% on paraphrases)
- Impact weights and scenario shocks are expert-set, not calibrated to history
- First-order stress maths: no convexity, gamma/vega or full loan revaluation
- Dictionary entity linking over a fixed fictional universe; one primary entity per text

Next steps

- Label real public news (with licences) and calibrate impact against market moves
- Fine-tune FinBERT / a small LLM for events; keep rules as an audit layer
- Historical scenario library (2008, 2020, 2022 rates) and full repricing
- Streaming ingestion (Kafka) with alerting and analyst feedback loop
- Coreference + entity resolution against reference data (LEI / tickers)

Benefit: headline-to-P&L in one explainable, offline workflow an analyst can audit. **Key limitation:** validated on synthetic data only; real-market calibration is the next step.